A full answer to Question 8.4: Fin $\otimes$ Fin separates ($\mathrm{LGBP_l}$) from ($\mathrm{LGBP_s}$)

Automated Mathematics Run `fa_banach_001`

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Status. Candidate full solution, likely valid; independent human review pending.

1 Source question

Sobota and Żuchowski ask the following in Question 8.4 on PDF page 47 of their preprint [1].

> Corollary 6.22 asserts that properties (LGBPs) and (LGBPl) are equivalent in the class of analytic P-ideals on ω. We are however not aware of any ideal on ω outside of this class which has only (LGBPl).
>
> **Question 8.4.** *Does there exist an ideal on ω with (LGBPl) but without (LGBPs)?*

In words: does there exist an ideal on ω which has the Local-to-Global Boundedness Property for lower semicontinuous submeasures ($\mathrm{LGBP_l}$), but not the corresponding property for arbitrary submeasures ($\mathrm{LGBP_s}$)?

2 Definitions

Let $\Omega = \omega \times \omega$, and write $R_n = \{n\} \times \omega$ and $A_n = \{k : (n, k) \in A\}$. We work with the standard ideal

$$\mathcal{J} = \mathrm{Fin} \otimes \mathrm{Fin} = \{A \subseteq \Omega : \{n : A_n \text{ is infinite}\} \text{ is finite}\}.$$

Since Ω is countable, any bijection $\omega \to \Omega$ transports $\mathcal{J}$ to an ideal on ω.

A submeasure is a monotone, subadditive map $\varphi : \mathcal{P}(\Omega) \to [0, \infty]$ which vanishes at the empty set and is finite on singletons. Put $\mathrm{Fin}(\varphi) = \{A : \varphi(A) < \infty\}$. An ideal $\mathcal{I}$ has ($\mathrm{LGBP_s}$) if every submeasure φ with $\mathcal{I} \subseteq \mathrm{Fin}(\varphi)$ is uniformly bounded on $\mathcal{I}$. It has ($\mathrm{LGBP_l}$) if every *lower semicontinuous* submeasure ψ with $\mathcal{I} \subseteq \mathrm{Fin}(\psi)$ satisfies $\psi(\Omega) < \infty$ [1].

3 Proof intuition

Counting the infinite rows gives exactly the non-lower-semicontinuous submeasure needed to destroy ($\mathrm{LGBP_s}$). Lower semicontinuity prevents this phenomenon: if an lsc submeasure were infinite on the whole grid but finite on $\mathcal{J}$, then every tail of rows would still have infinite value. One can choose a finite set of arbitrarily large value from each successive tail. Their diagonal union remains finite in every row, hence belongs to $\mathcal{J}$, but has infinite value—the required contradiction.

Theorem 1. *The ideal* $\mathcal{J} = \mathrm{Fin} \otimes \mathrm{Fin}$ *has* $(\mathrm{LGBP_l})$ *and does not have* $(\mathrm{LGBP_s})$*. Consequently, Question 8.4 has an affirmative answer in ZFC.*

Proof. For $A \subseteq \Omega$, set

$$S(A) = \{n \in \omega : A_n \text{ is infinite}\}, \qquad \rho(A) = |S(A)| \in \mathbb{N} \cup \{\infty\}.$$

This is a submeasure. Indeed, it is monotone, vanishes on finite sets, and

$$S(A \cup B) \subseteq S(A) \cup S(B),$$

because the union of two finite row sections is finite. Thus ρ is subadditive. Moreover, $\mathrm{Fin}(\rho) = \mathcal{J}$, while for $P_m = \bigcup_{n<m} R_n$ one has $P_m \in \mathcal{J}$ and $\rho(P_m) = m$. Hence ρ is finite but unbounded on $\mathcal{J}$, proving that $\mathcal{J}$ does not have $(\mathrm{LGBP_s})$.

Now let ψ be an lsc submeasure such that $\mathcal{J} \subseteq \mathrm{Fin}(\psi)$. Suppose for a contradiction that $\psi(\Omega) = \infty$. For every m, the initial row block P_m belongs to $\mathcal{J}$, so $\psi(P_m) < \infty$. With

$$T_m = \Omega \setminus P_m = \bigcup_{n \geq m} R_n,$$

subadditivity forces $\psi(T_m) = \infty$; otherwise $\psi(\Omega) \leq \psi(P_m) + \psi(T_m) < \infty$.

By lower semicontinuity, for every m there is a finite set $F_m \subseteq T_m$ such that $\psi(F_m) > m$. Let $A = \bigcup_m F_m$. Fix a row R_n. Since $F_m \subseteq T_m$, the set F_m misses R_n whenever $m > n$. Therefore

$$A \cap R_n \subseteq \bigcup_{m \leq n} F_m$$

is finite. Every row section of A is finite, so $A \in \mathcal{J}$. On the other hand, monotonicity gives $\psi(A) \geq \psi(F_m) > m$ for every m, hence $\psi(A) = \infty$, contradicting $\mathcal{J} \subseteq \mathrm{Fin}(\psi)$. Thus $\psi(\Omega) < \infty$, which is precisely $(\mathrm{LGBP_l})$. □

4 Checks, scope, and novelty

The witness ρ is not lower semicontinuous: finite approximations to a full row have value zero, while the full row has value one. Hence it does not conflict with the $(\mathrm{LGBP_l})$ conclusion. Also, $\mathcal{J}$ is not a P-ideal: the full rows R_n have no pseudounion in $\mathcal{J}$. The example therefore lies outside the analytic P-ideal class singled out immediately before the source question.

The run's lightweight indexes and a bounded web/arXiv search were queried by the source id, the exact LGBP terminology, $\mathrm{Fin} \otimes \mathrm{Fin}$, and close variants. The search found the source and general literature on this ideal, but no later answer to Question 8.4 or matching separation argument. This is not an exhaustive novelty claim.

Human-review focus. Check the two LGBP quantifiers, the lsc selection of the finite sets F_m, and the row-finiteness of their diagonal union.

References

[1] D. Sobota and T. Żuchowski, *The Nikodym and Grothendieck properties of Boolean algebras and rings related to ideals*, arXiv:2510.19744 (2025), Question 8.4.